
Course Syllabus (1-Year Timeline): Science Through Advanced Research, Learning, Application, and Building (STARLAB)

Instructor: Brian Hageman

Grade Level: 11th – 12th grade

Prerequisites: Completion of introductory science courses (Biology, Chemistry, or Physics) or instructor approval

Course Duration: 1 Year

Course Description

This course provides high school students with a unique opportunity to conduct rigorous, long-term scientific research in a chosen field of interest. With the support of a mentor-facilitator, students will design, execute, and report on self-directed research projects. Through these projects, students will acquire valuable skills in experimental design, data analysis, and scientific communication, creating a competitive portfolio for future academic and professional pursuits.

Course Objectives

By the end of this course, students will be able to:

- Develop and refine a scientific question or engineering problem.
 - Design and conduct an experiment or engineering solution with appropriate control and rigor.
 - Collect, analyze, and interpret data to make informed conclusions.
 - Present findings through written reports, visual media, and oral presentations.
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Units and Weekly Breakdown

Unit 1: Introduction to Scientific Research and Experimental Design (Weeks 1-2)

- **Topics Covered:**
 - Fundamentals of scientific inquiry and research ethics
 - Basics of experimental design: variables, controls, and replicability
 - Brainstorming and selecting research topics
- **Activities:**
 - Guided brainstorming sessions and literature review
 - Development of a research question and discussion with peers and mentors
- **Learning Outcomes:**
 - Students will identify a feasible, original research topic with a clearly defined question or hypothesis.

Unit 2: Project Planning and Hypothesis Development (Weeks 3-4)

- **Topics Covered:**
 - Developing testable hypotheses or engineering design criteria
 - Planning experimental procedures and identifying necessary materials
 - Safety and ethics, including form completion (e.g., IRB, safety forms)
- **Activities:**
 - Research planning workshop
 - Safety protocol review and approval processes (as required)
- **Learning Outcomes:**
 - Students will produce a research plan with a detailed timeline, materials list, and risk assessment if necessary.

Unit 3: Pilot Testing and Data Collection (Weeks 5-16)

- **Topics Covered:**
 - Pilot testing: refining methods, gathering preliminary data
 - Documentation of procedures and findings in a project journal
 - Techniques for systematic data collection and measurement
- **Activities:**
 - Conducting preliminary trials and adjusting experimental setup based on findings
 - Regular entries in a project journal, documenting observations and adjustments
- **Learning Outcomes:**
 - Students will conduct pilot tests, adjust their methodologies as needed, and begin systematic data collection.

Unit 4: Data Analysis and Interpretation (Weeks 17-20)

- **Topics Covered:**
 - Data organization, statistical analysis, and error assessment
 - Data visualization: charts, graphs, and other visuals for analysis
 - Comparison of findings to initial hypothesis or design goals
- **Activities:**
 - Statistical analysis workshop (basic to advanced methods, depending on project needs)
 - Data interpretation and discussion of preliminary findings
- **Learning Outcomes:**
 - Students will analyze and interpret their data, assessing whether results support the hypothesis or meet design criteria.

Unit 5: Scientific Reporting and Presentation Preparation (Weeks 21-28)

- **Topics Covered:**
 - Structure of scientific reports (introduction, methods, results, discussion, conclusion)
 - Creating visuals that accurately represent data
 - Techniques for effective oral presentations
- **Activities:**
 - Drafting and peer-reviewing research reports
 - Designing visual aids for project boards (graphs, tables, images)
 - Practice presentations with feedback sessions
- **Learning Outcomes:**
 - Students will produce a well-organized scientific report and prepare for a public presentation.

Unit 6: Project Presentation and Reflection (Weeks 29-34)

- **Topics Covered:**
 - Presentation delivery, engaging with an audience, and responding to questions
 - Reflection on the research process: challenges, solutions, and personal growth
 - Optional guidance on submitting projects to science fairs or journals
 - **Activities:**
 - Final presentation to peers, teachers, or at a school fair
 - Group reflection session on the course and individual projects
 - For interested students, preparation for submitting research to competitions or publications
 - **Learning Outcomes:**
 - Students will confidently present their findings, demonstrating understanding and articulation of their research.
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Assessment and Grading

Component	Weight
Project Proposal	15%
Research Journal/Documentation	20%
Mid-Project Progress Report	15%
Data Analysis and Interpretation	20%
Final Research Report	15%
Oral Presentation	15%

Resources and Tools

- **Primary Resources:** Access to lab equipment, relevant software for data analysis (e.g., Google Sheets, Excel, or statistical software), presentation software, and a research journal
 - **Supplementary Texts and Websites:**
 - *The Science Fair Project Guide* by The Society for Science & the Public
 - Online resources: SciFy Initiative, Rules Wizard (ISEF), local university databases
 - **Mentorship and Collaboration:** Regular access to mentor meetings, peer-review sessions, and guest speakers from scientific and engineering fields as available
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Expectations

Students are expected to:

- Engage actively in each phase of research, maintaining consistent documentation.
- Collaborate respectfully with peers and mentors, providing and receiving constructive feedback.
- Adhere to project deadlines and demonstrate a commitment to revising and refining their work based on data and peer insights.

Accommodations

Individualized support will be available to ensure accessibility for students with diverse needs. Adjustments in pacing, alternative materials, or additional mentor guidance will be provided as needed.